

PREPARED FOR

T6N, R4W, Mount Diablo Meridian

COORDINATES 38.3459°N 122.3047°W WGS84	ELEVATION 66 ft NAVD88 · MSL	STATE PLANE CA 2 (0402) · N 1,887,922.81 E 6,474,289.05 (US ft, NAD83)	FEMA ZONE Zone X 06055C0510FF	DECLINATION 13.0° E WMM 2026
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APN 038050009000 | ZONING AP | ACRES 0.481 ac

BEST GNSS WINDOW Mon, Jul 13 · 6:00 AM – 10:00 AM | PDOP 0.99 · 34 sv | NEAREST CORS
P261 · 14.1 mi ✓

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Site file · Google Earth
Download KMZ

RECORDER
Napa County Recorder-Clerk · 1127 First
St, Napa, CA 94559

NEAREST CORS
P261 · 14.1 mi

GENERATED
July 13, 2026

ASSESSOR
Napa County Assessor · 1127 First St,
Napa, CA 94559

NEAREST BM
JT9010 · 0.2 mi

SYSTEM
TopoScout · info@toposcout.com

This report compiles data from public and third-party sources (NGS, FEMA, USGS, NOAA, and county records) current as of the generation date shown above. GNSS planning values are predictive and conditions may change. This document is provided for planning purposes only and does not constitute a boundary survey, legal land description, or engineering certification. Verify all critical values against primary sources.

SITE & SIGNAL REPORT

COORDINATES	ELEVATION	MAGNETIC DECLINATION	FEMA FLOOD ZONE
38.3459, -122.3047	66 ft	13.0° E	X
WGS84 / NAD83	Above MSL (approx)	Compass reads 13.0° too far W	AREA OF MINIMAL FLOOD HAZARD · Panel 06055C0510FF

IONOSPHERIC CONDITIONS

PEAK KP INDEX

4.7

AVG KP

2.4

SOLAR FLUX F10.7

138 sfu

TEC (VTEC) ⚠ DEGRADED

19.0 TECU

-3.8 TECU vs median

ASSESSMENT

MODERATE ACTIVITY

Peak Kp 4.7 (avg 2.4). Solar flux F10.7 = 138 sfu. Moderate geomagnetic activity. Some ionospheric delay possible. Static sessions should use longer occupation times.

GNSS / DOP SUMMARY

DATE	AVG PDOP	BEST 4-HR WINDOW	AVG SATS	RATING
Mon, Jul 13	0.99	6:00 AM – 10:00 AM	34	Ideal

GPS + GLONASS + Galileo + BeiDou MEO · Full multi-constellation model · 15° elevation mask · Per-day independent computation
GNSS satellite geometry is accurate within a 50-mile radius of the entered location – orbital mechanics do not change meaningfully at ground-level distances. Weather data is specific to the entered coordinates.

🔥 WILDFIRE PROXIMITY

No Active Fires Within 150 Miles

No active fires as of report date (July 13, 2026) · Conditions may change – verify before fieldwork

✖ GPS CONSTELLATION HEALTH

⚠ 1 Satellite Currently Unhealthy – PRN 11

Active outage per NAVCEN · Affected satellites auto-excluded by receiver

Minor Obstruction

Low vegetation or sparse development nearby — minor sky obstruction possible.

Low Multipath

Minor reflective surfaces present — multipath generally manageable.

RECOMMENDED ELEVATION MASK

13° Minor vegetation — slight horizon reduction

✓ PDOP Check at 13°: 0.8–0.8 across survey window
PDOP remains ≤ 0.8 at 13° — geometry confirmed acceptable

5-POINT SAMPLE (CENTER + N/S/E/W · ~100 M)

Point	Cover Type	Obstruction	Multipath
Center	Dwarf Scrub	Low	Low
North	Dwarf Scrub	Low	Low
South	Dwarf Scrub	Low	Low
East	Dwarf Scrub	Low	Low
West	Dwarf Scrub	Low	Low

Dominant cover: Dwarf Scrub · Source: USGS NLCD 2024 via MRLC · 30 m resolution · Industry standard mask guidance per NGS field procedures

Monday, Jul 13

Overcast

9 mph

GO

High 93°F

Low 59°F

Wind max 9 mph

Gusts 12 mph

Precip 0.00"

Sunrise 5:56 AM

Sunset 8:33 PM

Waning Crescent 2%

GNSS: Avg PDOP **0.99** · Best 4-hr window 6:00 AM – 10:00 AM (PDOP 0.94, 34 sats avg)

GNSS / DOP - HOURLY

	PDOP	SATS	RATING
6:00 AM	0.84	36 sv	IDEAL
7:00 AM	0.95	34 sv	IDEAL
8:00 AM	0.95	33 sv	IDEAL
9:00 AM	1.00	33 sv	GOOD
10:00 AM	1.01	29 sv	GOOD
11:00 AM	1.11	25 sv	GOOD
12:00 PM	1.03	26 sv	GOOD
1:00 PM	1.08	26 sv	GOOD
2:00 PM	0.93	32 sv	IDEAL
3:00 PM	0.97	33 sv	IDEAL
4:00 PM	0.98	33 sv	IDEAL
5:00 PM	0.99	33 sv	IDEAL
6:00 PM	1.02	30 sv	GOOD
7:00 PM	0.99	32 sv	IDEAL



GO - GOOD CONDITIONS

Partly cloudy | Wind: 9.4 mph (gusts 11.6 mph) | Precip: 0" | Temp: 59°–93.2°F



DRONE WIND ANALYSIS



GO Peak: 9.4/11.6 mph S

* Sunrise: 5:56 AM Sunset: 8:33 PM fly window clipped to daylight

Best flight window: 6:00 AM – 9:00 PM (15 hrs)

Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≈turbulence	DIR
5 AM	2/4 mph		SE
6 AM	0/3 mph		W
7 AM	3/5 mph		E
8 AM	2/2 mph		SSW
9 AM	2/3 mph		SE
10 AM	4/5 mph		SSE
11 AM	5/5 mph		S
12 PM	5/6 mph		S
1 PM	5/5 mph		S
2 PM	9/10 mph		S
3 PM	7/8 mph		S
4 PM	8/9 mph		SSW
5 PM	9/11 mph		SSW
6 PM	9/11 mph		SSW
7 PM	9/12 mph		S
8 PM	8/11 mph		S

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ▶ FREE TO FLY

Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107

→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY](#)

Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

FIELD PLANNING REPORT

JOB: SG-MTGFG2 | SITE: N/A | LOCATION: 38.3459, -122.3047 (Napa County, CA)

REPORT DATE: Monday, July 13, 2026 | SURVEY START: Monday, July 13, 2026

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VERDICT: GO

Primary reason: PDOP geometry excellent (0.94–0.99 range), weather favorable (overcast, low wind, no precipitation), ionospheric activity moderate with manageable delays for static work.

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BEST SURVEY WINDOW

Monday, July 13, 2026: 0600–1000 hours local time

Rationale: Peak constellation geometry with 34 satellites available across all constellations. PDOP 0.94 is optimal. Overcast conditions reduce direct sun exposure for crew; winds remain light (max 9 mph). Window closes by 1000 as PDOP begins to degrade mid-morning.

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BEST 4-HOUR STATIC GNSS SESSION

0600–1000 hours, Monday, July 13

Recommended occupation: Minimum 60 minutes per point (extended from standard 30–45 min due to moderate Kp 4.7 and TEC estimate uncertainty). Vertical TEC 19.0 TECU is 3.8 TECU below 30-day median; ionospheric delays moderate but predictable in static post-processing. Multi-constellation receiver configuration (GPS+GLONASS+Galileo+BeiDou) strongly advised to mitigate PRN 11 (UNUSUFN) absence and maintain redundancy. Recommended elevation mask: 13° based on terrain. Use 15° operational mask per GNSS Planning input.

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DRONE OPERATIONS ASSESSMENT

Drone survey feasible. Overcast conditions reduce glint/reflectance; light winds (max 9 mph) acceptable for small–medium UAS. GNSS-assisted flight supported by excellent PDOP (0.99 avg). Recommend flight operations within 0600–1000 window to align with peak satellite geometry and favorable solar angle. Coordinate drone launches before 0930 to permit post-flight processing within daylight hours.

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FIELD SAFETY WARNINGS

- GPS PRN 11 currently unhealthy; receivers will auto-exclude. No field action required, but document in survey notes.
 - Air quality acceptable (AQI 45, Good). Standard PPE sufficient.
 - No active wildfires in proximity. All-clear for operations.
 - Temperature range 59–93°F; heat stress possible during midday. Ensure adequate hydration and shade scheduling during 1000+ hours (outside recommended window).
 - Dwarf scrub terrain (low multipath, low obstruction). Site is favorable; use recommended 13° elevation mask to minimize near-horizon noise.
- =====

SUMMARY

Monday, July 13, 2026, is a GO day for GNSS static survey and drone operations in Napa County. Optimal window is 0600–1000 local time, offering PDOP 0.94, 34-satellite constellation, and favorable weather (overcast, 9 mph wind max, no rain). Moderate ionospheric activity (Kp 4.7, TEC 19.0 TECU) warrants extended static occupation (60+ min/point) and multi-constellation receiver use. Terrain is benign (dwarf scrub, low multipath). GPS PRN 11 is unhealthy but auto-excluded; plan for 33-satellite geometry. Execute survey in the

recommended 0600–1000 window and defer non-critical operations to avoid afternoon heat stress.

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COUNTY RECORDING OFFICES

RECORDER / REGISTER OF DEEDS

Napa County Recorder-Clerk

1127 First St, Napa, CA 94559

ASSESSOR / TAX OFFICE

Napa County Assessor

1127 First St, Napa, CA 94559

CORS STATIONS – WITHIN 30 MILES

QR / LINK	STATION ID	NAME / LOCATION	DISTANCE
 NGS	P261	HUNTERHILLCN2004	14.1 mi
 NGS	P198	PETALUMAIRCN2004	17.5 mi
 NGS	OHLN	OHLONE PARK	23.5 mi
 NGS	P196	MEACHUMLFLCN2006	24.0 mi
 NGS	CASR	SANTA ROSA CA	24.9 mi
 NGS	P267	DIXONAVIATCN2005	26.3 mi

NGS BENCHMARKS – WITHIN 10 MILES

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
 Datasheet	JT9010	"13"	19.100m / 62.7ft NAVD88	Order	0.2 mi
 Datasheet	JT9018	"13 E"	17.970m / 59.0ft NAVD88	Order	0.2 mi
 Datasheet	JT9009	"13"	17.900m / 58.7ft NAVD88	Order	0.2 mi
 Datasheet	JT9017	"13 D"	17.890m / 58.7ft NAVD88	Order	0.2 mi
 Datasheet	JT9034	"20 A"	20.730m / 68.0ft NAVD88	Order	0.3 mi
 Datasheet	JT9012	"13"	19.060m / 62.5ft NAVD88	Order	0.4 mi
 Datasheet	JT9043	"25 A"	18.710m / 61.4ft NAVD88	Order	0.4 mi
	JT9016	"13 D"	18.200m / 59.7ft NAVD88	Order	0.5 mi

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
Datasheet					

FEMA FLOOD ZONE

X – AREA OF MINIMAL FLOOD HAZARD

Panel: 06055C0510FF

Effective: Sep 29, 2010

[View FEMA Map ↗](#)



DATA COLLECTOR IMPORT – NGS BENCHMARKS

NGS BENCHMARK CONTROL POINTS – Copy / Paste to Data Collector

Datum: NAD83 | Vertical: NAVD88 | Source: NGS NDE API

PID	NAME	LAT	LON	ELEV_FT	ORDER
JT9010	13	38.348333	-122.302778	62.7	
JT9018	13 E	38.348333	-122.302778	59.0	
JT9009	13	38.346667	-122.301111	58.7	
JT9017	13 D	38.345000	-122.301111	58.7	
JT9034	20 A	38.345000	-122.309444	68.0	
JT9012	13	38.351667	-122.304444	62.5	
JT9043	25 A	38.340000	-122.302778	61.4	
JT9016	13 D	38.341667	-122.297778	59.7	

[Download .txt File](#)

USGS STREAM GAUGES – within 25 miles

STATION	DIST	LEVEL	FLOW	FLOOD STATUS
NAPA R NR NAPA CA #11458000	1.5 mi	2.37 ft	4.39 cfs	● Normal
NATHANSON C A SONOMA CA #11458600	9.3 mi	—	—	● Normal
SONOMA C A AGUA CALIENTE CA #11458500	10.4 mi	8.84 ft	2.73 cfs	● Normal
NAPA R NR ST HELENA CA #11456000	14.1 mi	1.43 ft	3.23 cfs	● Normal
SONOMA CREEK A KENWOOD CA #11458433	14.9 mi	3.93 ft	0.19 cfs	● Normal
PUTAH C NR WINTERS CA #11454000	16.8 mi	7.85 ft	578 cfs	— Unknown
SUISUN SLOUGH NR BENICIA CA #11455740	19.6 mi	13.03 ft	2,230 cfs	— Unknown
POPE C A WALTER SPRINGS CA #11453590	21.1 mi	—	—	— Unknown

USGS LAKES & RESERVOIRS – within 150 miles

STATION	DIST	LEVEL	FULL POOL	LINK
CLEAR LK A LAKEPORT CA #11450000	58.1 mi	5.03 ft	N/R	USGS ↗

CALIFORNIA RESERVOIRS (DWR/CDEC) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
Lake Del Valle	60.6 mi	696.94 ft (-9 ft)	706 ft	36K af	67%
Folsom Lake	66.5 mi	448.16 ft (-18 ft)	466 ft	787K af	81%
Lake Oroville	93.4 mi	861.14 ft (-39 ft)	900 ft	2,858K af	81%
New Melones Lake	100.1 mi	1,020.15 ft (-68 ft)	1,088 ft	1,664K af	69%
Black Butte Lake	101.2 mi	—	228 ft	74K af	54%

USBR RESERVOIRS (Bureau of Reclamation) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
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ARMY CORPS RESERVOIRS (USACE) – within 150 miles

RESERVOIR	DIST	POOL ELEV	FULL POOL	STORAGE	% CAP
Del Valle	64 mi	696.9 ft	695.01 ft	—	100%
Camanche	89 mi	230.3 ft	221.86 ft	—	104%
Englebright Lake-Pool	95 mi	525.1 ft	522.35 ft	—	101%
Farmington Dam-Pool	99 mi	124.9 ft	117.09 ft	—	107%
Black Butte-Pool	101 mi	457 ft	458.36 ft	—	100%
Don Pedro	137 mi	—	N/R	—	—

USGS GROUNDWATER WELLS – within 15 miles

STATION	DIST	WELL DEPTH	WATER LEVEL	DATE	AQUIFER
005N004W32A002M	7 mi	200 ft	87.71 ft BGS	2014-11-05	N100CACSTL
005N004W28R001M	7 mi	51 ft	41.90 ft BGS	1977-10-03	N100CACSTL
004N004W04C001M	8 mi	80 ft	11.20 ft BGS	1977-10-03	N100CACSTL
004N004W05B001M	8 mi	—	54.10 ft BGS	1977-10-03	N100CACSTL
005N004W31R001M	8.1 mi	295 ft	—	—	—
004N004W05D002M	8.3 mi	60 ft	14.60 ft BGS	1977-10-03	N100CACSTL

004N004W06B001M	8.3 mi	201 ft	21.86 ft BGS	2022-06-07	—
004N004W02L001M	8.8 mi	—	9.90 ft BGS	1977-10-02	N100CACSTL

🕒 AI AQUIFER INTELLIGENCE — CLAUDE HAIKU

The Napa Valley Conglomerate/Sandstone/Limestone aquifer is a Pliocene-age sedimentary formation deposited in ancient marine and fluvial environments, now buried beneath alluvial cover. This formation acts as a primary and secondary water source for the region, with moderate to good productivity depending on local cementation and fracturing. Water availability at this site is moderate; the variable well depths (51–295 ft) and water levels (9.90–87.71 ft BGS) indicate that the aquifer is present but stratified, requiring proper well completion depth to tap consistently saturated zones.

FORMATION

Name	Napa Valley Conglomerate/Sandstone/Limestone
Age / Rock	Pliocene-age continental and shallow marine conglomerate, sandstone, and limestone
Extent	800–1200 sq mi (est. USGS/DWR)

DEPTH & RECHARGE

Depth Range	51–295 ft (5–30 floors)
Typical Depth	100–150 ft (10–15 floors)
Recharge Source	Precipitation on Mayacamas and Vaca Range outcrops; lateral groundwater flow from higher elevations; minor stream infiltration
Transit Time	15–40 years to 100 ft depth (est. USGS/DWR)

WATER QUALITY

pH	est. 7.0–7.6 (est. USGS/DWR)
Hardness	est. 250–450 mg/L CaCO3 (est. USGS/DWR)
TDS	est. 350–700 mg/L (est. USGS/DWR)
Notes	No sample data — typical for formation; local hardness and TDS elevated due to limestone and marine facies

TREND

Trend	Mixed trend: 1977 levels (9.90–54.10 ft BGS) suggest shallower water table; 2014 (87.71 ft BGS) and 2022 (21.86 ft BGS) show significant variability, consistent with drought/recovery cycles
Source	Well record water level measurements (1977, 2014, 2022); est. USGS/DWR regional drought impact data
Seasonal Swing	est. 10–25 ft (est. USGS/DWR); higher variation expected in drought years

Source: USGS National Water Information System · Search all wells in this area → USGS NWIS Mapper ↗ Water level data pending USGS API migration for CA state well numbers

LAND SUBSIDENCE & AQUIFER COMPACTION

✓ LOW SUBSIDENCE RISK — No known critically overdrafted basin

Source: CA DWR SGMA Bulletin 118 2024.1 · California only - full US pending server migration

🕒 AI SUBSIDENCE ASSESSMENT — CLAUDE HAIKU

The site is located in the North Coast Ranges aquifer system (N100CACSTL), an area with well-drained soils and no history of intensive groundwater overdraft. The coastal geology is characterized by consolidated bedrock and alluvial deposits with good vertical drainage, preventing the clay layer compression that typically drives subsidence in California's agricultural valleys. This location presents minimal subsidence risk for surveying purposes.

SUBSIDENCE ASSESSMENT

Susceptibility	Low
Primary Mechanism	None - consolidated terrain with adequate drainage
Annual Rate	<0.5 mm/yr (est. USGS)
Historic Loss	None documented
Aquifer Compaction	None - aquifer not over-pumped
SGMA Status	Not applicable - not in SGMA basin

BENCHMARK IMPLICATIONS

Monument Risk	Low - monuments stable
Elevation Validity	Good - no active subsidence
GNSS Recommendation	Standard benchmark verification adequate

SURVEY ACTION

Required Action	No special action required
Dataset	CA DWR SGMA Bulletin 118 2024.1
Coverage	California critically overdrafted basins only - full US coverage pending

STRUCTURAL & ENGINEERING PARAMETERS

SEISMIC (ASCE 7-22)

Site Class	D (default)
Risk Category	II

WIND / SNOW / ICE / FROST (ASCE 7-22)

Basic Wind Speed	92 mph
Ground Snow Load	5.8 psf
Winter Wind (W2)	0.35
Ice Thickness	None
Frost Depth	12 in minimum
Est. from IECC Zone 3C — AFI data unavailable	
Verify with local building official before construction	

CLIMATE ZONE (IECC 2021): Zone 3C · Marine · Marine - mild year-round, coastal influence (SF, LA coast) County: Napa

🕒 AI CLIMATE SUMMARY — RESIDENTIAL DESIGN

This marine climate stays mild year-round with moderate coastal winds (92 mph design), so your envelope can be lighter than cold climates—focus on air sealing and moderate insulation rather than extreme R-values, and plan for potential salt spray corrosion on metal details. Maximize south and west-facing glazing for passive solar gain during cooler months, but use shading devices (overhangs, screens) to prevent summer overheating on those same exposures, while keeping north-facing glass minimal to reduce heat loss. Your best passive strategy here is natural cross-ventilation through operable windows—design for strong prevailing breezes to cool the building during warm periods rather than relying heavily on air conditioning.

🕒 AI STRUCTURAL ASSESSMENT — CLAUDE HAIKU

This coastal marine site in northern California presents well-drained gravelly loam soils suitable for conventional shallow foundations, though seismic parameters are currently unavailable and may warrant confirmation with the local building official. Wind exposure and moderate snow loads are the primary climatic design drivers anticipated for this location, with standard detailing likely adequate pending seismic data review.

FOUNDATION

Type Conventional spread footing anticipated pending geotechnical investigation and seismic parameter confirmation

Basis Well-drained Coombs gravelly loam on 2–5% slopes suggests adequate bearing capacity for spread foundations; however, seismic design parameters (SDC, Ss, S1, SDS, SD1, PGA) are currently null and should be confirmed to assess if special detailing or deeper investigation is warranted

STRUCTURAL IMPLICATIONS

Seismic Seismic design parameters unavailable—consult engineer and local building official to confirm applicable Seismic Design Category and response spectrum values; detailing strategy may range from ordinary to special moment frames pending results

Wind Exposure Exposure C (open terrain with scattered obstructions) anticipated; 92 mph 3-sec gust wind speed (ASCE 7-22 Risk Category II) suggested for design—verify actual site exposure during field review

Wind Standard wind load design anticipated for Risk Category II; no hurricane exposure. Consult engineer to confirm exposure classification and applicable design pressures

Snow Moderate ground snow load of 5.8 psf anticipated with winter wind factor $W2 = 0.35$; standard roof design likely adequate—confirm roof geometry and exposure with structural engineer

INVESTIGATIONS & INSPECTIONS

Geotech Required Yes—geotechnical investigation recommended to confirm Site Class D assumption, bearing capacity, settlement estimates, and compliance with seismic parameter guidance from local building official

Liquefaction Screen Liquefaction screening recommended once seismic parameters and shallow groundwater depth are confirmed; well-drained soils and higher elevation suggest lower risk, but formal assessment advisable pending SDC determination

Special Inspections Anticipated scope depends on final Seismic Design Category determination; consult engineer and building official once parameters are confirmed

Code Edition ASCE 7-22 and IBC 2021 (planning reference only—confirm locally adopted edition)

Site Class Note Site Class D is a planning assumption based on soil type and drainage characteristics; licensed geotechnical engineer should confirm via subsurface investigation and shear-wave velocity testing or cone penetration resistance, particularly given seismic parameter uncertainty

SOILS — USDA SOIL SURVEY

MAP UNIT: Coombs gravelly loam, 2 to 5 percent slopes

SERIES: Coombs

DRAINAGE CLASS: Well drained

HYDRO GROUP: C

SOIL ORDER: Alfisols

SUBORDER: Xeralfs

🕒 AI SOIL INTELLIGENCE — CLAUDE HAIKU

The Coombs gravelly loam is a well-drained Alfisol (Xeralf) found on gently rolling upland terrain in the Coast Range and inland valleys of California. This soil formed in residuum and colluvium derived from mixed metamorphic and sedimentary rocks, typically over weathered bedrock at moderate depth. On-site, expect a gravelly surface layer overlying a clay-enriched substratum; the 2–5% slope gradient generally promotes good drainage, but the subsurface clay content and moderate permeability require attention during grading, foundation design, and any septic system evaluation.

Mixed metamorphic and sedimentary residuum and colluvium; upland sideslopes and ridges; scattered oak and chaparral or grassland vegetation.

DRAINAGE

Class	Well drained
Seasonal Saturation	None within 60 inches
Flooding Risk	Minimal on 2–5% slopes
Ponding Risk	None

ENGINEERING

Bearing Capacity	2,500–3,500 psf (shallow foundations)
Rating	Good
Shrink-Swell	Moderate — clay loam to clay substratum
Frost Action	Low
Cut Suitability	Good (gravel-rich surface layer; competent material)
Fill Suitability	Fair (clay substratum — moisture condition control required during compaction)

SEPTIC

Suitability	Conditional
Notes	Percolation test required; clay-rich substratum (15–35 cm depth) may restrict infiltration and limit long-term acceptance rates. Depth to limiting layer and seasonal moisture fluctuation should be verified.

HAZARDS

Excavation	Moderate — 15–35% surface gravel content and clay substratum; rock fragments may slow progress
Expansive Clay	None anticipated — clay content moderate; expansion potential low to moderate

Erosion	Low to Moderate on 2–5% slopes; increased risk if vegetative cover removed or drainage concentrated; gully initiation possible in disturbed headwater areas
Other	Moderate depth to bedrock (typically 40–60 inches) — verify before deep foundation or utility crossing design

Data sources: USGS National Water Information System · [waterdata.usgs.gov](#) · California DWR / CDEC · [cdec.water.ca.gov](#) · USBR RISE · [data.usbr.gov](#) · USACE CWMS · [water.usace.army.mil](#) · USGS NWIS Groundwater · [waterservices.usgs.gov](#) · USDA Web Soil Survey · [sdmdataaccess.nrcs.usda.gov](#) · CA DWR SGMA Bulletin 118 · [gis.water.ca.gov](#) · ASCE 7-22 Hazard Tool · [gis.asce.org](#) · USGS Design Maps · [earthquake.usgs.gov](#) · IECC 2021 Climate Zones · DOE Building America · Real-time data subject to revision

Water levels current as of report run: Jul 13, 2026, 2:36 AM UTC

* Live pool elevation not reported by this reservoir — full pool design elevation shown

 OIL & GAS WELLS California — within 10mi radius

8 total **0** active **7** plugged By status: 7 Plugged 1 Idle

By type: 8 DH

Lease / Well	Status	Type	Operator	Depth	Dist
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	7.2 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	7.2 mi
McKenzie-Hazard Syndicate	Plugged	DH	McKenzie-Hazard Syndicate	—	7.4 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	8.6 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	8.8 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	9.5 mi
Capell Oil Corp.	Idle	DH	Capell Oil Corp.	—	9.7 mi
Cordelia Oil & Gas Co.	Plugged	DH	Cordelia Oil & Gas Co.	—	12 mi

AI Assessment — Well Proximity

Well proximity is not a planning concern for this site. The nearest well is a plugged Mobil Oil Corporation dry hole located 7.2 miles away, well beyond any actionable review distance. No active producers, injection wells, or other well types exist within relevant distances to affect surface rights, easements, or environmental assessment.

Source: California public well records. Active = currently producing or permitted. Plugged = abandoned/decommissioned. Data updated nightly.

200
total in search area

10
within 5 mi

108
within 25 mi

Site Name	Status	Commodities	Distance
Roderick Sand and Gravel Pit	Occurrence	Sand/Gravel	1.6 mi
Basalt Rock Co	Producer	Clay	2 mi
Cicero Pumice Quarry	Occurrence	PUM	2.9 mi
Mount George King Pumice Deposit.	Occurrence	PUM	2.9 mi
Basalt Rock Co.	Occurrence	DIT	3.1 mi
Mt George Pumice Deposit	Producer	PUM	3.1 mi
Wilson Pumice Quarry	Occurrence	PUM	3.7 mi
Unknown	Occurrence	Mercury	4.4 mi
Silva Pumice Quarry	Occurrence	PUM	4.7 mi
Silva Pumice Quarry	Producer	PUM	5 mi
Walker Pumice Quarry	Occurrence	PUM	5.1 mi
Walker Pumice Quarry	Producer	PUM	5.1 mi
Mountain	Occurrence	Mercury	5.4 mi
Unknown	Occurrence	Mercury	7.6 mi
Unknown	Occurrence	Stone/Crushed	7.8 mi
Unnamed Location	Past Producer	Sand/Gravel	8.5 mi
Unnamed Location	Past Producer	Sand/Gravel	8.5 mi
Unnamed Prospect	Occurrence	Chromium	8.8 mi
Unnamed Location	Occurrence	Chromium	9 mi
Unknown	Occurrence	Chromium	9.1 mi
Bella Oak	Past Producer	Mercury	9.4 mi
Unknown	Occurrence	MG	9.7 mi
Serres Pit	Past Producer	Sand/Gravel	10.4 mi
Unnamed Location	Unknown	Sand/Gravel	10.5 mi
Conn Valley	Occurrence	Manganese	10.6 mi

Showing 25 of 200 deposits. Full dataset via USGS MRDS.

AI Assessment — Mining Context

Three active producers operate within 5 miles of the site. The closest is Basalt Rock Co (clay operation, 2 miles), followed by Mt George Pumice Deposit (3.1 miles) and Silva Pumice Quarry (5 miles). Expect heavy truck traffic during operating hours, dust generation from aggregate/pumice extraction, and noise from crushing or processing operations at these facilities.

Source: USGS Mineral Resources Data System (MRDS). Status: Producer = active/recent; Past Producer = historical; Prospect = explored not mined; Occurrence = mineralization noted.

ENDANGERED & THREATENED SPECIES - USFWS Critical Habitat within 25mi

6
Endangered

6
Threatened

12
Total w/ Critical Habitat

6 Endangered species with designated critical habitat. ESA Section 7 consultation may be required for federal nexus projects.

COMMON NAME	SCIENTIFIC NAME	STATUS
California tiger Salamander	Ambystoma californiense	Endangered

COMMON NAME	SCIENTIFIC NAME	STATUS
Conservancy fairy shrimp	<i>Branchinecta conservatio</i>	Endangered
Contra Costa goldfields	<i>Lasthenia conjugens</i>	Endangered
Soft bird's-beak	<i>Cordylanthus mollis</i> ssp. <i>mollis</i>	Endangered
Suisun thistle	<i>Cirsium hydrophilum</i> var. <i>hydrophilum</i>	Endangered
Vernal pool tadpole shrimp	<i>Lepidurus packardii</i>	Endangered
Alameda whipsnake (=striped racer)	<i>Masticophis lateralis euryxanthus</i>	Threatened
California red-legged frog	<i>Rana draytonii</i>	Threatened
Delta smelt	<i>Hypomesus transpacificus</i>	Threatened
Northern spotted owl	<i>Strix occidentalis caurina</i>	Threatened
Vernal pool fairy shrimp	<i>Branchinecta lynchi</i>	Threatened
western snowy plover	<i>Charadrius nivosus nivosus</i>	Threatened

AI SPECIES & ESA ASSESSMENT - CLAUDE HAIKU

ESA Section 7 Implications Summary The site's location within critical habitat for 12 federally listed species triggers ESA Section 7 consultation requirements if federal permitting, funding, or authorization is involved. The most planning-relevant species are California tiger salamander and vernal pool species (tadpole shrimp, fairy shrimp) due to their dependence on seasonal wetland habitats common to the region, followed by California red-legged frog and Delta smelt if hydrological connectivity exists. The recommended next step is to conduct a jurisdictional wetland delineation and habitat assessment to determine the presence of occupied/suitable habitat on-site, which will establish consultation necessity and inform project design modifications (such as avoidance, buffers, or compensatory mitigation). Concurrent coordination with USFWS and CDFW is essential to clarify federal trigger applicability and species-specific survey protocols. --- **For planning purposes only. Consult a licensed environmental biologist for formal ESA compliance.**

Source: USFWS Critical Habitat Final Designated Areas - FWS Open Data - Critical habitat does not equal species range

NATIVE PLANTS - iNaturalist Research-Grade within 10mi

57 5051 2026-07-04
Unique Species Total Observations Most Recent

COMMON NAME	SCIENTIFIC NAME	LAST OBS
American black nightshade	<i>Solanum americanum</i>	2026-05-29
Bearded Clover	<i>Trifolium barbigerum</i>	2026-05-15
bird's foot cliffbrake	<i>Pellaea mucronata</i>	2026-04-28
Blow Wives	<i>Achyrachaena mollis</i>	2026-05-20
blue oak	<i>Quercus douglasii</i>	2026-05-09
Bolander's Phacelia	<i>Phacelia bolanderi</i>	2026-04-27
box elder	<i>Acer negundo</i>	2026-05-18
Braun's Giant Horsetail	<i>Equisetum telmateia braunii</i>	2026-06-22
California bay	<i>Umbellularia californica</i>	2026-06-24
California beeplant	<i>Scrophularia californica</i>	2026-05-20
California black oak	<i>Quercus kelloggii</i>	2026-05-15
California buckeye	<i>Aesculus californica</i>	2026-06-02
California Dutchman's Pipe	<i>Aristolochia californica</i>	2026-05-26
California milkwort	<i>Rhinotropis californica</i>	2026-05-15
California mugwort	<i>Artemisia douglasiana</i>	2026-06-02
California poppy	<i>Eschscholzia californica</i>	2026-06-24
California Wild Rose	<i>Rosa californica</i>	2026-05-09
Canyon liveforever	<i>Dudleya cymosa</i>	2026-05-30
-	<i>Castilleja densiflora densiflora</i>	2026-05-09
clay mariposa lily	<i>Calochortus argillosus</i>	2026-05-15
coast live oak	<i>Quercus agrifolia</i>	2026-05-13
coastal woodfern	<i>Dryopteris arguta</i>	2026-05-20
common selfheal	<i>Prunella vulgaris</i>	2026-06-22
Diogenes' lantern	<i>Calochortus amabilis</i>	2026-05-15
distant phacelia	<i>Phacelia distans</i>	2026-05-15
-	<i>Downingia concolor concolor</i>	2026-05-15
Dwarf Brodiaea	<i>Brodiaea terrestris</i>	2026-05-15
Elegant Clarkia	<i>Clarkia unguiculata</i>	2026-05-16
goldback fern	<i>Pentagramma triangularis</i>	2026-05-20
Ithuriel's Spear	<i>Triteleia laxa</i>	2026-06-15
maroonspot calicoflower	<i>Downingia concolor</i>	2026-05-15

COMMON NAME	SCIENTIFIC NAME	LAST OBS
narrowleaf milkweed	<i>Asclepias fascicularis</i>	2026-06-25
narrowleaf mule-ears	<i>Wyethia angustifolia</i>	2026-05-15
Narrowleaf Willow	<i>Salix exigua</i>	2026-06-24
needleleaf navarretia	<i>Navarretia intertexta</i>	2026-05-15
northern California black walnut	<i>Juglans hindsii</i>	2026-06-24
orange bush monkeyflower	<i>Diplacus aurantiacus</i>	2026-05-29
Oregon Ash	<i>Fraxinus latifolia</i>	2026-06-02
Pacific Bleeding Heart	<i>Dicentra formosa</i>	2026-05-04
Pacific madrone	<i>Arbutus menziesii</i>	2026-07-04
Pacific poison oak	<i>Toxicodendron diversilobum</i>	2026-06-02
pineapple-weed	<i>Matricaria discoidea</i>	2026-05-20
Redwood Sorrel	<i>Oxalis smaliana</i>	2026-05-28
small-headed clover	<i>Trifolium microcephalum</i>	2026-05-15
Superb Mariposa Lily	<i>Calochortus superbus</i>	2026-06-11
thimble clover	<i>Trifolium microdon</i>	2026-05-15
tomcat clover	<i>Trifolium willdenovii</i>	2026-05-15
Toyon	<i>Heteromeles arbutifolia</i>	2026-06-19
trailing blackberry	<i>Rubus ursinus</i>	2026-06-24
Tree Clover	<i>Trifolium ciliolatum</i>	2026-05-15
valley oak	<i>Quercus lobata</i>	2026-06-24
wavy-leafed soap plant	<i>Chlorogalum pomeridianum</i>	2026-05-20
western sycamore	<i>Platanus racemosa</i>	2026-05-17
white brodiaea	<i>Triteleia hyacinthina</i>	2026-05-15
white-tipped clover	<i>Trifolium variegatum</i>	2026-05-15
Winecup Clarkia	<i>Clarkia purpurea</i>	2026-05-15
Yellow Mariposa Lily	<i>Calochortus luteus</i>	2026-06-11

AI PLANT COMMUNITY SUMMARY - CLAUDE HAIKU

Ecological Assessment: 38.3459, -122.3047 This site supports a **mixed oak woodland and grassland community**, dominated by drought-tolerant species like blue oak, California black oak, and California buckeye interspersed with native grassland forbs and spring ephemerals. The high diversity of clay-loving indicators (clay mariposa lily, Castilleja) and moisture-responsive species (Braun's Giant Horsetail, California bay) suggests variable microtopography with seasonal wet areas and well-drained slopes typical of the northern California foothills. Soils are likely clay-enriched in swales and depressions with moderate to poor drainage during winter/spring, transitioning to very dry conditions in summer—meaning grading and drainage work should preserve seasonal seeps and avoid compaction in low-lying areas to maintain native plant viability. The predominance of deep-rooted oak woodland species indicates that surface disruption and drainage modifications could significantly impact both immediate vegetation and longer-term hydrologic function. **Consult a licensed botanist for site-specific vegetation assessment.**

Source: iNaturalist research-grade observations - Native species filter applied

FIELD HAZARDS - Venomous & Toxic Species within 25mi

635	154	91	1439
Venomous Snakes	Venomous Spiders	Stinging Insects	Toxic Plants
VENOMOUS SNAKES			
SPECIES		OBSERVATIONS	
Crotalus oreganus oreganus		635 obs	
VENOMOUS SPIDERS			
SPECIES		OBSERVATIONS	
Latrodectus hesperus		154 obs	
STINGING INSECTS			
SPECIES		OBSERVATIONS	
Dasymutilla aureola		91 obs	
Dasymutilla californica		91 obs	
Dasymutilla sackenii		91 obs	
TOXIC PLANTS			
SPECIES		OBSERVATIONS	
Toxicodendron diversilobum		1100 obs	
Datura wrightii		57 obs	

SPECIES	OBSERVATIONS
<i>Datura stramonium</i>	57 obs
<i>Datura ferox</i>	57 obs
<i>Conium maculatum</i>	199 obs
<i>Urtica gracilis</i>	83 obs
<i>Urtica gracilis holosericea</i>	83 obs

AI FIELD SAFETY BRIEFING - CLAUDE HAIKU

Field Safety Brief – Northern California Survey Area **Highest-Priority Hazards:** Your greatest risks are toxic plants—particularly poison oak (*Toxicodendron diversilobum*), which dominates this region with 1,439 observations—followed by Northern Pacific rattlesnakes (*Crotalus oreganus oreganus*, 635 observations) and black widow spiders (*Latrodectus hesperus*). *Datura* species and poison hemlock add secondary plant hazards. **Practical Avoidance:** Wear long sleeves, pants, and closed-toe boots; learn to identify poison oak (three leaflets, variable color) and avoid touching it or brushing equipment against it. Watch your foot placement in tall grass and brush for rattlesnakes, especially during warm hours; make noise while walking. Check equipment and clothing regularly for spiders. Wash contaminated gear separately to prevent oil transfer. Bring a first-aid kit and know the location of nearest emergency services. **Seasonal Considerations:** Rattlesnake activity peaks April through October; late summer heat increases snake encounters. Poison oak causes more severe reactions during warmer months when foliage is dense. Late fall and winter offer safer conditions but reduced visibility in brush. **Always verify current conditions locally before fieldwork.**

Source: iNaturalist research-grade observations - Always verify locally before fieldwork

FCC MICROWAVE PATHS — Licensed beam paths crossing within 1320ft of site

3	545ft	3
Beam Paths	Closest Beam	Fixed P2P

TX CALL	RX CALL	TYPE	PATH MI	CLOSEST
WBO61	WBO59	Fixed Point-to-Point	—	545ft
KYH37	WQAT627	Fixed Point-to-Point	—	927ft
WRCR329	WQAT627	Fixed Point-to-Point	—	1289ft

Source: FCC ULS microwave license database — PostGIS spatial query — Paths within 1320ft (0.25mi) of site — Towers may be 50+ miles away — FCC Part 101 licensed paths

FAA AIRWAYS — ATS Routes crossing site area

26	13	2	11
Total Airways	Jet Routes	Victor Airways	Other Routes

AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
J1	Jet Route	High	18,000 ft	—	303.8°	121.4°
J110	Jet Route	High	18,000 ft	—	137.9°	318.2°
J126	Jet Route	High	18,000 ft	—	325.5°	144.0°
J143	Jet Route	High	18,000 ft	—	325.0°	145.8°
J3	Jet Route	High	18,000 ft	—	342.8°	161.8°
J32	Jet Route	High	18,000 ft	—	19.2°	199.6°
J501	Jet Route	High	18,000 ft	—	317.1°	135.4°
J58	Jet Route	High	18,000 ft	—	65.3°	245.9°
J65	Jet Route	High	18,000 ft	—	304.1°	121.1°
J80	Jet Route	High	18,000 ft	—	65.3°	245.9°
J84	Jet Route	High	18,000 ft	—	50.8°	231.0°
J88	Jet Route	High	18,000 ft	—	308.0°	127.3°
J94	Jet Route	High	18,000 ft	—	65.3°	245.9°
Q1	Q-Route	High	—	—	334.5°	154.1°
Q120	Q-Route	High	—	—	38.1°	219.8°
Q122	Q-Route	High	—	—	37.8°	219.6°
Q124	Q-Route	High	—	—	37.8°	219.6°
Q138	Q-Route	High	—	—	47.7°	229.0°
Q3	Q-Route	High	—	—	164.0°	344.6°
Q5	Q-Route	High	—	—	161.4°	341.7°
T257	T-Route	Low	—	—	321.1°	140.5°
T259	T-Route	Low	—	—	41.6°	221.8°
T261	T-Route	Low	—	—	31.4°	211.7°
T263	T-Route	Low	—	—	308.8°	128.2°
V107	Victor Airway	Low	7,000 ft	—	295.4°	114.2°
V108	Victor Airway	Low	4,500 ft	—	117.7°	296.8°

RAILROAD PROXIMITY — FRA North American Rail Network

8
Rail Lines

OWNER / OPERATOR	TYPE	TRACKS	PASSENGER	STRACNET	SUBDIVISION
CFNR	Other	1	No	No	—
CFNR	Yard	1	No	No	—
CFNR	Mainline	1	No	No	VALLEJO
SMRT	Mainline	1	No	No	BRAZOS JCT.
NVRR	Mainline	1	No	No	MARTINEZ
USG	Other	1	No	No	—
CFNR	S	1	No	No	—
NVRR	Yard	1	No	No	—

Source: FRA North American Rail Network (NARN) — Updated July 2025 — STRACNET = Strategic Rail Corridor Network (DOD)

WILDLIFE & NATURAL HISTORY - iNaturalist within 25mi

200 90 65 5 14 6
Total Observations Total Species Birds Mammals Reptiles Amphibians

BIRDS (65 species)

COMMON NAME	Scientific Name	OBS
Wild Turkey	<i>Meleagris gallopavo</i>	10
Snowy Egret	<i>Egretta thula</i>	7
California Quail	<i>Callipepla californica</i>	5
Western Bluebird	<i>Sialia mexicana</i>	5
California Towhee	<i>Melospiza crissalis</i>	5
Great Egret	<i>Ardea alba</i>	4
Red-tailed Hawk	<i>Buteo jamaicensis</i>	4
Black Phoebe	<i>Sayornis nigricans</i>	4
Northern Flicker	<i>Colaptes auratus</i>	3
Cooper's Hawk	<i>Astur cooperii</i>	3
Ash-throated Flycatcher	<i>Myiarchus cinerascens</i>	3
Western Screech-Owl	<i>Megascops kennicottii</i>	3
Turkey Vulture	<i>Cathartes aura</i>	3
Mallard	<i>Anas platyrhynchos</i>	3
Bushtit	<i>Psaltiriparus minimus</i>	3

+ 50 additional species

MAMMALS (5 species)

COMMON NAME	Scientific Name	OBS
Black-tailed Jackrabbit	<i>Lepus californicus</i>	3
Coyote	<i>Canis latrans</i>	2
Mule Deer	<i>Odocoileus hemionus</i>	2
Eastern Fox Squirrel	<i>Sciurus niger</i>	1
Striped Skunk	<i>Mephitis mephitis</i>	1

REPTILES (14 species)

COMMON NAME	Scientific Name	OBS
Western Fence Lizard	<i>Sceloporus occidentalis</i>	10
Pacific Gopher Snake	<i>Pituophis catenifer catenifer</i>	7
Northern Pacific Rattlesnake	<i>Crotalus oreganus oreganus</i>	6
California King Snake	<i>Lampropeltis californiae</i>	3
Pond Slider	<i>Trachemys scripta</i>	2
Western Skink	<i>Plestiodon skiltonianus</i>	2
Ring-necked Snake	<i>Diadophis punctatus</i>	1
Common Garter Snake	<i>Thamnophis sirtalis</i>	1

COMMON NAME	Scientific Name	OBS
Western Yellow-bellied Racer	Coluber constrictor mormon	1
Pacific Ringneck Snake	Diadophis punctatus amabilis	1
Northwestern Pond Turtle	Actinemys marmorata	1
Common Sharp-tailed Snake	Contia tenuis	1
North American Racer	Coluber constrictor	1
Southern Alligator Lizard	Elgaria multicarinata	1

AMPHIBIANS (6 species)

COMMON NAME	Scientific Name	OBS
Pacific chorus frog	Pseudacris regilla	5
American Bullfrog	Lithobates catesbeianus	3
Foothill Yellow-legged Frog	Rana boylei	3
Western Toad	Anaxyrus boreas	2
Rough-skinned Newt	Taricha granulosa	1
California Giant Salamander	Dicamptodon ensatus	1

AI WILDLIFE & ECOLOGICAL ASSESSMENT - CLAUDE HAIKU

Ecological Character Assessment The site at 38.3459, -122.3047 (Napa County, California) supports a diverse wildlife community characteristic of oak woodland and grassland habitat with proximity to riparian or freshwater resources, as evidenced by the abundance of wetland-associated birds (egrets), aquatic reptiles (pond turtles, pond sliders), and moisture-dependent amphibians (Pacific chorus frogs, California giant salamanders, rough-skinned newts). Several species present indicate regulatory considerations: the Foothill Yellow-legged Frog and California Giant Salamander are California Species of Special Concern requiring sensitive habitat management, while the Northwestern Pond Turtle is a California Department of Fish and Wildlife (CDFW) sensitive species. The presence of both native reptile diversity and introduced species (Eastern Fox Squirrel, American Bullfrog, Pond Slider) suggests moderate disturbance with intact structural habitat; however, the community composition indicates this site likely contains seasonal or year-round water features that warrant jurisdictional assessment under California Fish and Game Code and Clean Water Act provisions. Development review should prioritize preservation of riparian corridors and wetland complexes, with botanical surveys to assess oak woodland quality and potential rare plant communities. **For planning purposes only. Consult a qualified biologist for site-specific biological assessments.**

Source: iNaturalist research-grade observations — Data reflects community science observations and may not represent complete species inventory

✳️ PASSIVE + ACTIVE SOLAR PLANNING — Lat 38.35°

1 • AVERAGE SUN HOURS

Period	Peak Hrs/Day	Notes
Annual average	6.6 hrs	Overall solar resource
Summer average	9 hrs	June–August
Winter average	3.9 hrs	December–February
Worst month (Dec)	3.2 hrs	Size battery storage to this

2 • IDEAL SOLAR ARRAY PITCH

Optimal tilt = 38.3° (latitude) → 9/12 pitch

Pitch	Angle	% of Optimal	Notes
4/12	18.4°	97%	good
5/12	22.6°	97.6%	good
6/12	26.6°	98.2%	↪ excellent
7/12	30.3°	98.8%	↪ excellent
8/12	33.7°	99.3%	↪ excellent
9/12	36.9°	99.8%	↪ annual optimal
10/12	39.8°	99.8%	↪ excellent
12/12	45°	99%	↪ excellent

Going from 5/12 to 6/12 at framing costs ~\$800–1,200 one time. On a 10kW system that's +723 kWh/yr (~\$108 value/yr). Over 25yr panel life: ~\$2710 total gain.

3 • SOLAR HEAT GAIN — BTU/HR PER SQ FT

Use these values as inputs for eQUEST, EnergyPlus, Manual J, or HERS rating

Surface	Winter	Summer	Design Note
South glass (SHGC 0.4)	51	31	Minimize south glass
South glass (SHGC 0.6)	76	46	Clear/uncoated glass
North glass (any)	10	13	Negligible — diffuse only
East / West glass	27	104	⚠ Minimize — afternoon load
Roof — dark (absorb 0.95)	236	301	Highest load surface
Roof — light (absorb 0.35)	87	111	Cool roof — significant reduction
Roof — metal (absorb 0.25)	62	79	Best roof option for heat
South wall — dark	94	69	
South wall — light	31	23	Light color = major savings
Hardscape — asphalt	181	301	⚠ Heat island — avoid near structure
Hardscape — concrete	124	206	Also reflects onto adjacent walls
Hardscape — light pave	67	111	Recommended within 50ft
Pool surface	23	38	Low absorb + evaporative cooling

Concrete reflection onto south wall: ~4 BTU/hr·ft² additional load per 100sf of adjacent hardscape.
Pool specular reflection: angle-dependent — east/west pools can cause morning/afternoon glare into glass.

4 • ROOF PITCH SOLAR LOAD — BTU/HR PER 100 SQ FT (SUMMER)

Pitch	Dark Roof	Light Roof	Reduction vs Flat
0/12	301 BTU	111 BTU	baseline (worst)
4/12	241 BTU	89 BTU	-20% dark / -20% light
6/12	196 BTU	72 BTU	-35% dark / -35% light
8/12	166 BTU	61 BTU	-45% dark / -45% light
12/12	135 BTU	50 BTU	-55% dark / -55% light

Scale: multiply BTU × (your roof sf ÷ 100). Pitch + light color = biggest single cooling move.

5 • THERMAL TIME LAG — WALL ASSEMBLY

Peak outdoor temp ~15:00. Heat arrives at interior = peak + lag hours.

Assembly	Lag	Peak Interior Load	Note
2x4 frame (R-15)	0.6 hrs	~15:36 PM	R-value only — no mass benefit
2x6 frame (R-23)	0.9 hrs	~15:54 PM	R-value only — no mass benefit
2x8 frame (R-30)	1.2 hrs	~16:12 PM	R-value only — no mass benefit
2x10 frame (R-37)	1.5 hrs	~16:30 PM	good
2x12 frame (R-44)	1.8 hrs	~16:48 PM	good
ICF 6" core	7 hrs	~22:00 PM	✓ shifts past sunset — natural vent window
8" CMU block	6 hrs	~21:00 PM	✓ shifts past sunset — natural vent window
12" concrete	8.5 hrs	~23:30 PM	✓ shifts past sunset — natural vent window

In 3C Marine: ICF or CMU shifts peak load past sunset. Open windows, flush heat naturally. AC load drops significantly.

6 • TREE SHADOW PLANNING — TREE DUE SOUTH, 100FT FROM STRUCTURE

Tree Ht	Shadow Reaches	Summer hrs/day	Equinox hrs/day	Winter hrs/day	WUI note
10ft	All year	1.1	1	1.2	OK
20ft	All year	2.1	1.9	2.4	OK
30ft	All year	3.1	2.9	3.7	Zone 2 check
40ft	All year	4	3.8	5.2	△ verify
50ft	All year	4.8	4.6	7.3	△ verify
60ft	All year	5.6	5.5	9.3	✗ Zone 1

Deciduous tree: winter solar penetration ~85% (leaves off) — best of both worlds in hot-dry climates.

Evergreen tree: winter penetration ~15% — good windbreak, poor solar strategy south of structure.

△ WUI zones: tree placement must be coordinated with defensible space requirements. Verify with local fire authority — WUI boundaries vary by jurisdiction.

7 • ATTIC VENTING — PER 100 SQ FT OF ROOF AREA

Standard	Unbalanced (1/150)	Balanced† (1/300)	Notes
Code minimum	8 sq in	4 sq in	Moisture protection only
Thermal comfort	24 sq in	12 sq in	Keeps attic within 20°F of outdoor
WUI mesh 1/16"	60 sq in	30 sq in	2.5× compensation — mesh clogs
WUI intumescent	24 sq in	12 sq in	Self-closing — no compensation needed

Pitch factor: 0/12×1.0 · 4/12×0.80 · 6/12×0.65 · 8/12×0.55 · 12/12×0.45

Color factor: Dark×1.0 · Medium×0.75 · Light×0.55 · Metal×0.45

Scale: multiply sq in × (your roof sf ÷ 100) × pitch factor × color factor

†**Balanced** = ≥40% NFA within top third of roof height (ridge) + ≥40% within bottom third (soffit)

WUI conflict: Soffit vents are ember collection points. In high WUI zones — eliminate soffit vents, use ridge-only (unbalanced/1/150). In low WUI — balanced system preferred.

△ Verify WUI zone with local fire authority — boundaries vary significantly by jurisdiction.

🏠 AI SOLAR & PASSIVE DESIGN SUMMARY — HOMEOWNER GUIDE

Your Home's Solar & Energy Plan You're in a sweet spot for solar—think of your location as getting about 6 and a half hours of "prime charging time" for panels every day on average, with even longer hours in summer when you need cooling help. Your roof's angle is nearly perfect already (around a 9:12 pitch), so if you ever add solar panels, they'd love that slope. Here's what most people miss: your dark roof is essentially a heat collector working against you in summer, soaking up the sun like a black car parked in the driveway—painting it light colored could cut that heat gain by two-thirds, which saves real money on air conditioning. Your west-facing windows are the real culprit though; that afternoon sun hammering in from the west is brutal in this climate, so heavy curtains or external shading there will make the biggest single difference in comfort and cooling costs. The good news is a tall deciduous tree on your south side would block about four hours of summer sun when you need it most, then drop its leaves in winter to let the warm sun in—it's basically free air conditioning and heating working together. One simple way to think about thermal mass: it's like a sponge for heat, where thicker, heavier walls (like ICF) slowly absorb excess heat during the day and release it at night, while a standard wood frame house is like a thin cloth that heats up and cools down quickly with the sun. These numbers are planning estimates—your architect or energy modeler can plug them directly into their software for a full analysis.

Planning estimates only — clear-sky model, standard assumptions. Values formatted for eQUEST / EnergyPlus / Manual J input. Verify with licensed architect, energy modeler, or solar installer before specifying.



Technical Brief — for architects, engineers, and surveyors

Site Intelligence Briefing 38.34586°N, 122.30466°W | Napa County, California

Structural Design Drivers & Climate Envelope

This site falls within IECC Zone 3C (Marine) with an Annual Freezing Index of 0°F-days, indicating no meaningful frost cycling and mild winter thermal loads. Wind and snow design parameters are not available in your dataset, but the marine climate classification and zero AFI strongly suggest that thermal and precipitation loads are modest compared to inland or high-elevation California sites. Without seismic data in your briefing materials, do not assume this Napa County location is exempt from seismic design; confirm IBC seismic design category and spectral response parameters through USGS interactive tools or your structural engineer before finalizing the lateral system. The marine microclimate may reduce wind demand relative to ridge or valley sites in the same county—verify with local building official or wind consultant.

Geotechnical Foundation Strategy

The site is mapped on Coombs gravelly loam (2–5% slopes) classified as Hydro Group C with well-drained characteristics. Group C soils typically have moderate infiltration rates and are suitable for conventional shallow foundations; the well-drained nature eliminates immediate concerns about saturated conditions or liquefaction potential in the upper bearing strata. Soil texture classification is not provided in your data, so request a Phase I or preliminary geotechnical report to confirm bearing capacity, settlement potential, and any cobble/boulder content within the gravelly loam matrix—this will inform footing depth and width. Because frost depth is not recorded (consistent with AFI = 0°F-days), frost-induced heave is not a design driver; footings may be set shallower than in colder zones, but establish minimum depth through local building code and geotechnical engineer recommendations.

Hazard Assessment

No Quaternary faults are mapped within the search radius of this location, and subsidence basin data are absent, reducing those specific risks. However, this does not eliminate seismic hazard—Napa County is seismically active. Obtain site-specific seismic parameters (peak ground acceleration, spectral accelerations, and soil amplification factors) from a seismic hazard map or Phase I ESA update. Wildfire risk data are not included in your briefing; given the county location and proximity to wine country terrain, confirm local fire hazard severity zone (FHSZ) classification and any defensible-space or ignition-resistant construction requirements with the Napa County planning or fire marshal's office.

Site Context & Mining Proximity

Three active mining operations are within 5 miles of this site: Basalt Rock Co. (CLY, 2 mi), Mt George Pumice Deposit (PUM, 3.1 mi), and Silva Pumice Quarry (PUM, 5 mi). Blasting, dust, vibration, and traffic from these operations may affect site noise, air quality, and ground vibration during construction and operations. Additionally, 200 MRDS (Mineral Resources Data System) deposits are recorded within roughly 35 miles, indicating the region's mineral extraction legacy; confirm whether the site itself or nearby subsurface may have been prospected or mined historically. Eight oil/gas wells exist within 10 miles, though none are currently active or producing—verify plugging status and any subsurface pressure/gas migration risks through California Division of Oil, Gas and Geothermal Resources records.

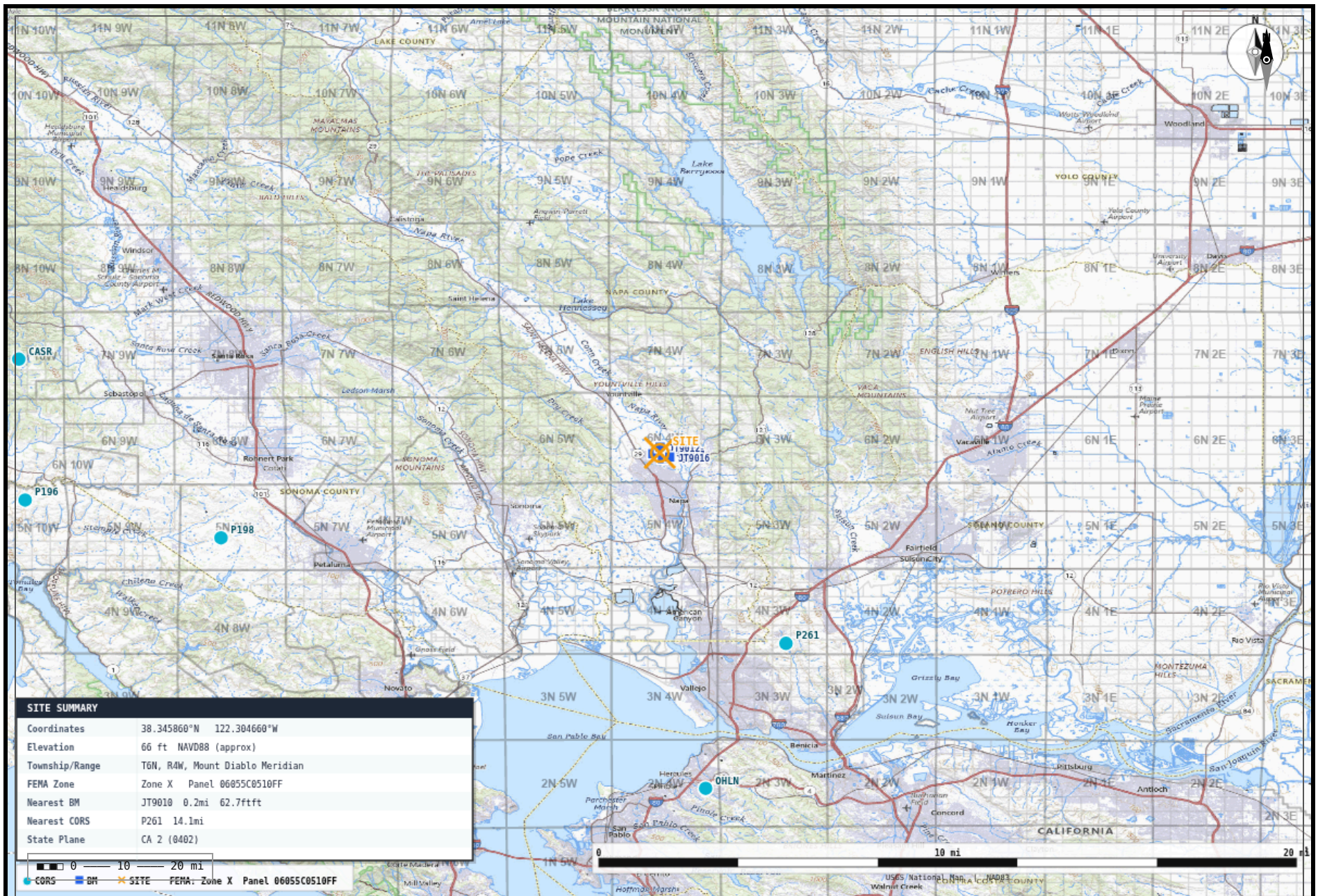
Bottom Line

Obtain seismic design parameters and local wind/snow data before finalizing structural loads, commission a Phase I geotechnical and Phase I ESA (including historical mining use and well status), and verify fire hazard zone classification and mining operation impacts with Napa County planning.



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Planning aid only — not a substitute for licensed professional review, geotechnical investigation, or code compliance verification.

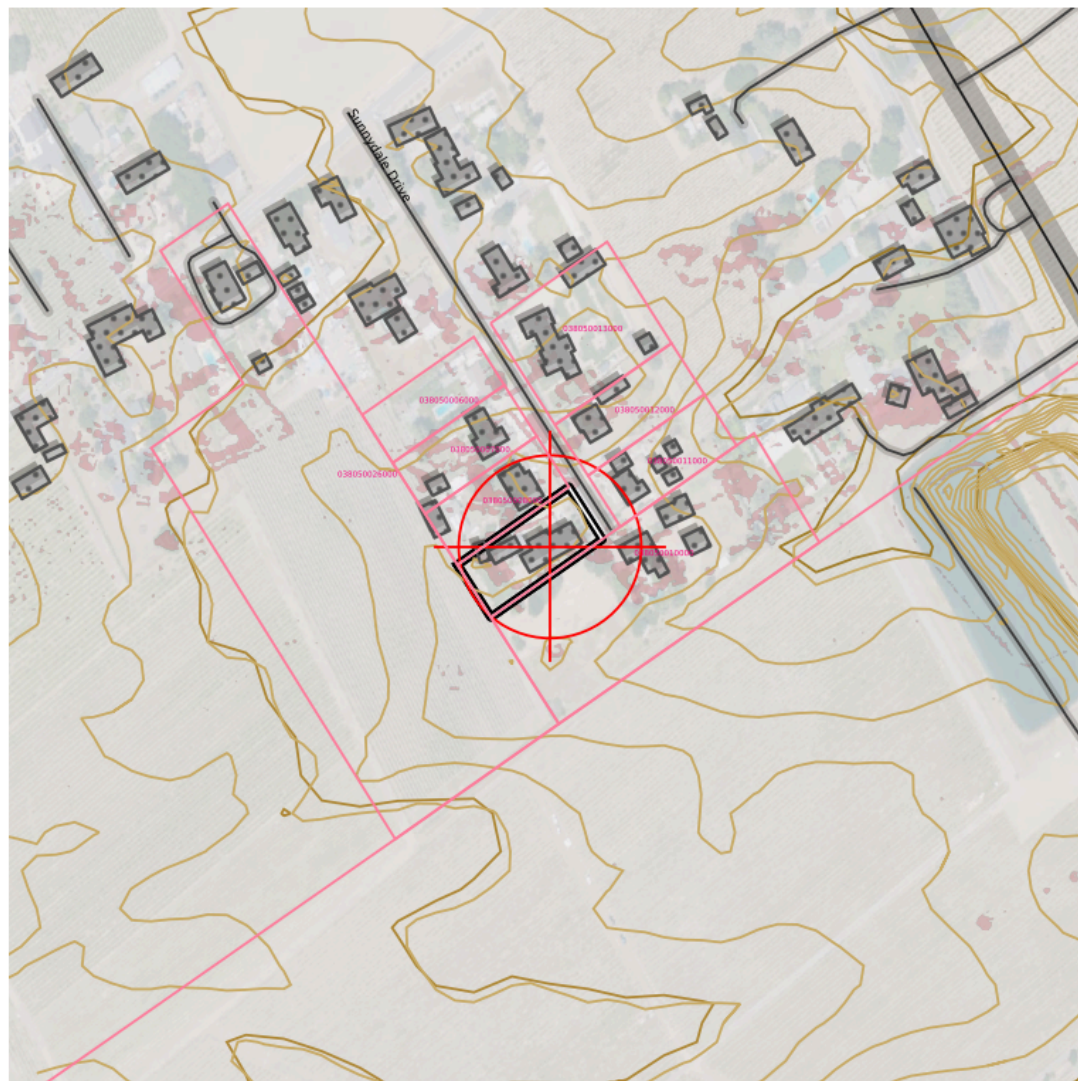


TOPOSCOUT
FIELD MAP

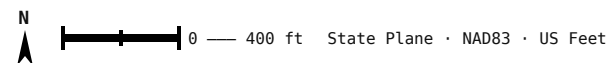
TopoScout Field Map
Lat: 38.34586 Lng: -122.30466
38.345860°N 122.304660°W | Radius: 30 mi

Generated: July 13, 2026
Base: USGS National Map
Generated by TopoScout - outkastdesigns.com

TOPOSCOUT SITE PLAN PREVIEW | \$19 UPGRADE FOR AUTOCAD DXF FILE



INCLUDED LAYERS



- Contours (Index/Major/Minor/1ft) ■ PLSS Section/Township/QQ ■ Roads Major/Local/Trail + ROW ■ Buildings ■ Hydro ■ Power Lines ■ Pipeline ■ Railroad ■ Benchmarks + Labels
- County Boundary ■ OTLS Grid ■ Site Info ■ Elevation Labels ■ 3D Polylines (all geometry)

✂ FIELD RESOURCE LINKS

PRE-MISSION PLANNING

Trimble GNSS Planning Online –

<https://trimblegnssplanningonline.com/>

Satellite visibility, DOP charts, sky plots

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Drone airspace authorization, TFRs, restricted areas

Aloft (Kittyhawk) – <https://www.aloft.ai/>

Field-friendly drone airspace map and LAANC authorization

NOAA Space Weather — Live Kp –

<https://www.swpc.noaa.gov/products/planetary-k-index>

Real-time Kp index, geomagnetic storm alerts

POST-PROCESSING

NGS OPUS – <https://geodesy.noaa.gov/OPUS/index.jsp>

Online Positioning User Service – static GNSS processing

NGS OPUS-RS – <https://geodesy.noaa.gov/OPUSI/index.jsp>

Rapid static processing – sessions as short as 15 minutes

CORS Data Download – [https://geodesy.noaa.gov/CORS/cors-](https://geodesy.noaa.gov/CORS/cors-data.shtml)

[data.shtml](https://geodesy.noaa.gov/CORS/cors-data.shtml)

Raw RINEX observation files from CORS stations

FEMA & FLOOD

FEMA Flood Map Service Center –

<https://msc.fema.gov/portal/home>

FIRM panels, flood zone lookup by address or coordinates

FEMA Elevation Certificate – [https://www.fema.gov/flood-](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

[insurance/work-with-nfip/elevation-certificate](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

Current EC form, instructions, LOMA/LOMR submission guidance

BASE STATION & NETWORK

NOAA CORS Network – [https://geodesy.noaa.gov/CORS/cors-](https://geodesy.noaa.gov/CORS/cors-data.shtml)

[data.shtml](https://geodesy.noaa.gov/CORS/cors-data.shtml)

Continuously operating reference stations, data download

NGS CORS Station Map –

<https://geodesy.noaa.gov/CORS/GoogleMap/CORSmap.shtml>

Interactive map of all CORS stations with status

IGS Station Monitor – <https://www.igs.org/network>

International GNSS Service – global base station health

REFERENCE & CALCULATIONS

NGS Datasheets – [https://www.ngs.noaa.gov/cgi-](https://www.ngs.noaa.gov/cgi-bin/datasheet.prl)

[bin/datasheet.prl](https://www.ngs.noaa.gov/cgi-bin/datasheet.prl)

Benchmarks, control points, passive monuments

NGS Coordinate Conversion (NCAT) –

<https://geodesy.noaa.gov/NCAT/>

NADCON5, datum transforms, state plane conversions

NOAA Magnetic Declination –

<https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml>

Declination calculator for any location and date

NGS Geoid Models – <https://geodesy.noaa.gov/GEOID/>

GEOID18 and current model downloads, GEOID calculator

DRONE OPERATIONS

FAA Part 107 — Commercial Rules –

https://www.faa.gov/uas/commercial_operators

Full Part 107 regulatory summary for commercial UAS operations

FAA DroneZone – <https://faadronezone.faa.gov/>

Drone registration, operator certificates, waiver management

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Pre-flight airspace check, TFRs, LAANC authorization

FAA Part 107 Waivers –

https://www.faa.gov/uas/commercial_operators/part_107_waivers

Waiver applications for operations beyond standard 107 rules

Aloft — Field Airspace Map – <https://www.aloft.ai/>

Mobile-friendly real-time airspace, LAANC, flight logging

AIR QUALITY & WILDFIRE SMOKE

[AirNow Fire & Smoke Map](https://www.fire.airnow.gov)

<https://www.fire.airnow.gov>

Current fire locations, smoke plumes, PM2.5 readings by location

[AirNow Interactive Map](https://gispub.epa.gov/airnow)

<https://gispub.epa.gov/airnow>

Real-time AQI monitors, hourly NowCast, 48-hr forecast overlays

[NOAA Smoke Forecast](https://airquality.weather.gov)

<https://airquality.weather.gov>

48-hr smoke concentration forecast, HRRR-Smoke model

